American Computer Science League

2023-2024 ● Contest 1: Short Problems ● Senior Division

1. Computer Number Systems

If the tickets to an event were numbered sequentially in base 16 and the first one sold was $3A5_{16}$ and the last one sold was $F4A_{16}$, how many tickets were sold expressed in base 10?

A. 2982
B. 2882
C. 2981
D. 2992
E. 2892

2. Computer Number Systems

What is the value of $7245_8 * 17_8$ expressed in base 8?

A. 165553
B. 155653
C. 156653
D. 156553
E. 157553

3. Recursive Functions

Find $f(f(f(15)))$ given:

$$f(x) = \begin{cases} (f(x-2)) \bmod 4 + 2 & \text{if } x \geq 10 \\ (f(x-1)) \bmod 5 + 4 & \text{if } 6 < x < 10 \\ x + 5 & \text{if } x \leq 6 \end{cases}$$

A. 2
B. 4
C. 5
D. 8
E. 11

4. Recursive Functions

Ann decided to make a 96" square quilt**.** The square was divided into 9 equal squares. The center square of each was left alone and the surrounding squares were divided into 9 equal squares. This process continued until the 9 squares formed measured less than 2 inches on a side. How many of the smallest squares were made?

A. 5832
B. 5184
C. 4608
D. 4096
E. 576

5. What Does This Program Do?

What is the output of the following program?

```
for yr = 2021 to 2025
    num = yr
    s1 = 0
    s2 = 0
    while num != 0
        d = num % 10
        s1 = s1 + d
        s2 = s2 + d^2
        num = int(num/10)
    end while
    if yr%s1 == 0 && yr%s2 == 0 then
        output yr
    end if
next yr
```

A. 2021
B. 2022
C. 2023
D. 2024
E. 2025